

Urban-attribute results across splits

Qwen3.5-9B (per-city & seen/unseen) — 4-way image ablation

Setup & what each split tests

Two LoRA-trained **Qwen3.5-9B** models, each evaluated on **its own validation split** and on the **shared benchmark**. Four image modes: **blind** (no images), **sv** (street-view only), **sat** (marked satellite only), **full** (both).

- **per-city val** — *unseen locations within cities the model trained on*. Tests in-distribution generalisation. ($n = 6,984$)
- **seen/unseen val** — *entirely held-out cities the model has **never seen at all***. The hard test: geographic generalisation to brand-new cities. ($n = 8,831$)
- **benchmark** — a stratified **10% of locations from every city** (BENCHMARK_RATIO=0.10, location-level), extracted *before* either split, so it is identical and held-out for both models. ($n = 5,240$)

Fusion = full – max(sv, sat). All: thinking off, greedy, edge 512 (matches training). Bold sv/sat = better single view.

How to read fusion on these 9 tasks

The 9 urban-attribute tasks are, **by construction, answerable from a single view** — land use, building height, road type, etc. are each readable from *either* the satellite or the street-view alone. So we **expect** $\text{full} \approx \text{the better single view}$, and $\text{fusion} \approx 0$ here is the **designed behaviour, not a model deficiency**.

What these tables actually show:

- **full never loses** to the better single view ($\text{fusion} \gtrsim 0$ throughout) — adding the second view doesn't confuse the model.
- Where one view is uninformative, full correctly tracks the **informative** one (bolded sv/sat per row).
- The **mismatch_*** tasks (appendix) are **literally unanswerable from one view**: the other view *is the answer choices*. There, fusion is +40–70 — the mechanism works whenever the task actually forces it.

per-city — OWN validation split (seen cities, new locations)

task	blind	sv	sat	full	full—best1
Land use	68.6	76.6	74.5	75.2	-1.4
Building height	41.0	64.8	67.0	66.3	-0.8
Urban density	55.1	62.0	54.7	59.4	-2.7
Junction type	58.0	68.3	74.1	74.6	+0.4
Green space	100.0	100.0	100.0	100.0	+0.0
Amenity richness	30.3	57.6	54.7	62.4	+4.8
Road type	66.3	76.5	71.3	75.2	-1.2
Road surface	94.2	94.0	93.7	94.2	+0.3
Transit density	28.5	46.9	47.6	50.6	+3.0
OVERALL (all topics)	48.6	58.4	54.3	75.5	

seen/unseen — OWN validation split (never-seen cities)

task	blind	sv	sat	full	full – best1
Land use	65.9	69.3	68.2	69.1	-0.1
Building height	40.4	59.2	55.1	58.8	-0.4
Urban density	70.1	66.8	66.4	70.0	+3.1
Junction type	68.9	71.4	72.0	73.5	+1.5
Green space	100.0	99.1	98.7	96.0	-3.1
Amenity richness	26.1	54.1	48.7	54.0	-0.1
Road type	65.8	72.0	70.1	70.0	-2.0
Road surface	98.8	98.1	97.2	98.1	+0.0
Transit density	27.2	42.9	44.2	45.2	+0.9
OVERALL (all topics)	48.6	55.7	52.4	73.1	

These cities appear *nowhere* in training — accuracy here is pure geographic generalisation.

Both models on the SHARED benchmark — 9 tasks

task	per-city full	per-city fus	seen/uns full	seen/uns fus
Land use	75.3	-0.2	73.9	+1.4
Building height	66.8	-0.5	66.8	+0.5
Urban density	65.1	+2.4	74.6	+3.1
Junction type	75.1	+1.8	76.3	+2.4
Green space	100.0	+0.0	98.0	-1.5
Amenity richness	58.7	+2.1	55.6	+0.0
Road type	76.5	-1.0	76.7	+0.0
Road surface	95.7	+0.7	95.0	-0.3
Transit density	50.1	+4.5	49.4	+3.6

Mean fusion over 9 tasks — per-city **+1.1**, seen/unseen **+1.0** (excl. leaked green-space: +1.2 / +1.3). Disjoint cities + locations from both training sets.

per-city — benchmark (all 4 modes)

task	blind	sv	sat	full	full—best1
Land use	68.6	75.1	75.5	75.3	-0.2
Building height	45.3	66.3	67.4	66.8	-0.5
Urban density	54.6	62.7	61.8	65.1	+2.4
Junction type	61.4	68.6	73.4	75.1	+1.8
Green space	100.0	100.0	100.0	100.0	+0.0
Amenity richness	26.4	56.5	53.4	58.7	+2.1
Road type	68.6	77.4	72.4	76.5	-1.0
Road surface	95.7	93.4	95.0	95.7	+0.7
Transit density	28.3	45.6	43.7	50.1	+4.5
OVERALL (all topics)	48.5	57.7	54.9	75.9	

seen/unseen — benchmark (all 4 modes)

task	blind	sv	sat	full	full–best1
Land use	63.4	72.4	72.0	73.9	+1.4
Building height	50.5	66.3	63.7	66.8	+0.5
Urban density	53.2	64.8	71.5	74.6	+3.1
Junction type	62.0	71.3	74.0	76.3	+2.4
Green space	100.0	99.5	99.0	98.0	-1.5
Amenity richness	25.7	55.6	53.0	55.6	+0.0
Road type	68.6	76.7	69.6	76.7	+0.0
Road surface	95.7	95.4	95.0	95.0	-0.3
Transit density	26.4	44.9	45.8	49.4	+3.6
OVERALL (all topics)	48.3	58.3	54.9	75.9	

External RS-VLM baselines — GeoChat & SkySenseGPT (zero-shot)

Two off-the-shelf remote-sensing VLMs (LLaVA-1.5-7B forks, CLIP-ViT-L/14 @ 504px) evaluated on the **shared benchmark, satellite-only, zero-shot**. They are **single-image, overhead-only** — so they see *only* the marked satellite tile (our sat mode), never street-view, on all 9 Family-1 tasks ($n = 3,135$).

- They **never trained on the red-dot marker** nor on our attribute taxonomies — this is a pure out-of-distribution transfer probe.
- Fair column to compare against is **our trained sat** on the same benchmark (single satellite view, like-for-like input).

All: greedy, max_new_tokens=8, 0% unparseable. SkySenseGPT > GeoChat overall.

Zero-shot RS-VLMs vs. our trained sat (benchmark, sat-only)

task	GeoChat	SkySenseGPT	pc sat	su sat
Land use	48.7	53.4	75.5	72.0
Building height	30.5	34.7	67.4	63.7
Urban density	13.5	14.5	61.8	71.5
Junction type	8.4	30.8	73.4	74.0
Green space	50.7	53.7	100.0	99.0
Amenity richness	13.5	12.6	53.4	53.0
Road type	10.7	18.1	72.4	69.6
Road surface	75.6	80.2	95.0	95.0
Transit density	21.4	26.1	43.7	45.8
OVERALL (9 tasks)	27.8	33.4	54.9	54.9

pc sat / su sat = our per-city / seen-unseen Qwen3.5-9B, sat mode, same benchmark. Zero-shot RS models trail our trained sat by **~21–27 pts overall**; closest on **road surface** (texture, view-agnostic), near chance on count/context tasks (urban density, amenity richness, junction type).

Appendix — ALL 14 tasks (full & fusion, every run)

task	pc-val	fus	su-val	fus	pc-bench	fus	su-bench	fus
Land use	75.2	-1.4	69.1	-0.1	75.3	-0.2	73.9	+1.4
Building height	66.3	-0.8	58.8	-0.4	66.8	-0.5	66.8	+0.5
Urban density	59.4	-2.7	70.0	+3.1	65.1	+2.4	74.6	+3.1
Junction type	74.6	+0.4	73.5	+1.5	75.1	+1.8	76.3	+2.4
Green space	100.0	+0.0	96.0	-3.1	100.0	+0.0	98.0	-1.5
Amenity richness	62.4	+4.8	54.0	-0.1	58.7	+2.1	55.6	+0.0
Road type	75.2	-1.2	70.0	-2.0	76.5	-1.0	76.7	+0.0
Road surface	94.2	+0.3	98.1	+0.0	95.7	+0.7	95.0	-0.3
Transit density	50.6	+3.0	45.2	+0.9	50.1	+4.5	49.4	+3.6
Camera direction	46.5	+21.4	44.7	+20.8	42.3	+17.3	43.7	+20.9
Mismatch bin (easy)	95.2	+48.1	96.6	+47.1	97.4	+44.7	98.8	+46.1
Mismatch bin (hard)	90.4	+39.2	87.8	+37.6	90.7	+44.9	87.9	+43.7
Mismatch MCQ (easy)	95.4	+69.2	95.0	+70.8	96.2	+69.8	92.6	+67.9
Mismatch MCQ (hard)	84.3	+57.2	82.7	+57.6	86.0	+62.7	84.8	+55.1

pc=per-city, su=seen/unseen, val=own split, bench=shared benchmark. **mismatch_*** fuse strongly (+40–70)
 — they *require* both views by construction.